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United States Patent	12398411
Kind Code	B2
Date of Patent	August 26, 2025
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Method for producing selenoneine

Abstract

The purpose of the present invention is to provide a method for producing selenoneine that allows production of selenoneine at higher yields, even if an inorganic selenium compound is used as a selenium compound. This purpose can be achieved by a method for producing selenoneine, comprising the step of applying histidine and a selenium compound to a transformant to obtain selenoneine, wherein the transformant has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes.

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Appl. No.:	17/251975
Filed (or PCT Filed):	June 14, 2019
PCT No.:	PCT/JP2019/023590
PCT Pub. No.:	WO2019/240243
PCT Pub. Date:	December 19, 2019

Prior Publication Data

Document Identifier	Publication Date
US 20230159968 A1	May. 25, 2023

Foreign Application Priority Data

JP	2018-114919	Jun. 15, 2018
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Publication Classification

Int. Cl.: C12P17/10 (20060101); C12N9/10 (20060101); C12N15/52 (20060101); C12N15/70 (20060101); C12N15/80 (20060101); C12P9/00 (20060101); C12P13/04 (20060101)

U.S. Cl.:

CPC C12P17/10 (20130101); C12N9/1025 (20130101); C12N9/1085 (20130101); C12N15/52 (20130101); C12N15/70 (20130101); C12N15/80 (20130101); C12P9/00 (20130101); C12P13/04 (20130101); C12Y203/0103 (20130101); C12Y205/01047 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This is the US National Stage of International Patent Application No. PCT/JP2019/023590, filed Jun. 14, 2019, which claims the benefit of priority to Japanese Patent Application No. 2018-114919, filed Jun. 15, 2018, the disclosure of which is incorporated herein by reference in its entirety.

BRIEF DESCRIPTION OF THE DESCRIBED SEQUENCES

(2) The nucleic and/or amino acid sequences provided herewith are shown using standard letter abbreviations for nucleotide bases, and one letter code for amino acids, as defined in with 37 CFR 1.831 through 37 CFR 1.835. Only one strand of each nucleic acid sequence is shown, but the complementary strand is understood as included by any reference to the displayed strand. The Sequence Listing is submitted as an .txt file named 96217_303_4_seq.txt, approximately 40,000 bytes, created Dec. 22, 2020, the contents of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(3) The present invention relates to a method for producing selenoneine. In particular, the present invention relates to a method for producing selenoneine using a microorganism having the ability to produce selenoneine.

BACKGROUND ART

(4) Selenoneine is an organic selenium compound that is known to have antioxidant activity and the ability to promote cell growth. The application of selenoneine to the prevention and treatment of diseases associated with selenium(Se) deficiency is expected.

(5) Known methods for producing selenoneine include extraction of selenoneine from organs or blood of animals (See, Patent Document 1), and use of fission yeast *Schizosaccharomyces pombe* transformed with genes involved in ergothioneine biosynthesis (See, Non-Patent Document 1).

(6) A method for producing selenoneine is known, which includes a step of applying histidine and a selenium compound to a transformant overexpressing the inserted EgtA gene (See, Patent Document 2).

CITATION LIST

Patent Document

(7) Patent Document 1: JP5669056 Patent Document 2: WO2017/026173

Non-Patent Document

(8) Non-Patent Document 1: PLoS One 2014 May 14; 9(5): e97774

SUMMARY OF INVENTION

Technical Problem

(9) The methods described in Patent Document 1 and Non-Patent Document 1 cannot produce selenoneine at higher yields. These methods are unsuitable for industrial-scale production of selenoneine.

(10) In contrast, the method described in Patent Document 2 is enable to produce selenoneine at higher yields as compared to the methods described in Patent Document 1 and Non-Patent Document 1. The method described in Patent Document 2 allows for industrial-scale production of selenoneine.

(11) The organic selenium compounds such as selenocystine, which is used in the method described in Patent Document 2, are expensive. Therefore, the method of using the conversion of the organic selenium compounds to selenoneine is uneconomical. An inorganic selenium compound may be converted to selenoneine instead of the organic selenium compound. However, if even selenious acid is used as an inorganic selenium compound in the method described in Patent Document 2, a sufficient amount of selenoneine is not produced.

(12) Accordingly, it is an objective of the present invention to provide a method for producing selenoneine that allows production of selenoneine at higher yields, even if an inorganic selenium compound is used as a selenium compound.

Solution to Problem

(13) To find solutions to the above-described problems, the present inventors attempted to produce selenoneine in large quantities by adding selenious acid as a substrate to a culture medium in which fungal cells were grown in order to efficiently produce selenoneine using selenious acid.

Specifically, the method described in Patent Document 2 was modified as follows: using a (AsEgtA+AsEgtC) transformant obtained using *Aspergillus sojae* as a host organism; culturing the transformant in a medium suitable for *Aspergillus sojae* for 2 days; adding selenious acid into the culture medium; and then further culturing. However, the production of selenoneine hardly increased.

(14) It was believed that this is because *Aspergillus sojae* is sensitive to selenious acid. Therefore, as long as selenious acid is used, it is difficult to produce selenoneine in large quantities.

(15) Accordingly, the present inventors have repeated trial and error concerning the use of another inorganic selenium compound as a starting material in place of selenious acid. As a result, the present inventors have finally focused on a biosynthetic pathway of sulfur assimilation.

Specifically, if the first biosynthetic pathway for converting sulfuric acid to hydrogen sulfide and the second biosynthetic pathway for converting hydrogen sulfide to cysteine are used, it was believed that selenoneine would be possible to be produced by using selenic acid and its salts instead of sulfuric acid.

(16) The present inventors decided to use MetR in the first biosynthetic pathway, and SatA and CysB in the second biosynthetic pathway. MetR promotes the reaction of converting sulfuric acid to sulfurous acid and further converting sulfurous acid to hydrogen sulfide. SatA catalyzes the reaction that converts serine to acetylserine. CysB catalyzes the reaction that converts acetylserine and hydrogen sulfide to cysteine. The present inventors have prepared a transformant that overexpresses a MetR gene, a SatA gene and/or a CysB gene in addition to an EgtA gene. The present inventors have subjected selenic acid to the transformant to successfully produce selenoneine.

(17) Surprisingly, the addition of selenic acid to the culture medium, which increased the amount of fungal cells of the transformant, allowed for a mass production of selenoneine. More surprisingly, the using of the transformant overexpressing the EgtA gene, the MetR gene, the SatA gene and the CysB gene allowed for a high selenoneine content in the resulting selenoneine extracts.

(18) Accordingly, the present inventors have finally succeeded in creating a method for producing selenoneine in large quantities from selenic acid by using the transformant that overexpresses the gene encoding the enzyme that catalyzes the reaction that converts selenic acid to selenoneine, in addition to the EgtA gene. It is these successful examples and findings that ultimately led to the completion of the present invention.

(19) According to one embodiment of the present invention, methods and transformants of [1] to [10] below are provided.

(20) [1] A method for producing selenoneine, including the step of applying histidine and a selenium compound to a transformant to obtain selenoneine, wherein the transformant has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes.

(21) [2] The method according to [1], wherein the transformant has two copies to eight copies of the EgtA gene inserted therein and can overexpress the inserted gene.

(22) [3] A method for producing selenoneine, including the step of applying histidine and a selenium compound to a transformant to obtain selenoneine, wherein the transformant has two copies to eight copies of an EgtA gene inserted therein and can overexpress the inserted gene.

(23) [4] The method according to any one, of [1] to [3], wherein the selenium compound includes at least one selenium compound selected from the group consisting of selenic acid, selenious acid, selenium chloride, selenium tetrachloride, selenium, selenium dioxide, selenides, selenium sulfide, dimethylselenium, selenophosphate and salts thereof.

(24) [5] The method according to any one of [1] to [4] wherein a host organism of the transformant is a microorganism expressing at least one enzyme selected from the group consisting of selenic acid reductase, selenocysteine lyase and serine dehydratase.

(25) [6] The method according of any one of [1] to [4] wherein a host organism of the transformant is at least one microorganism selected from the group consisting of microorganisms of genus *Aspergillus*, genus *Escherichia*, genus *Trichoderma*, genus *Fusarium*, genus *Penicillium*, genus *Rhizopus* and genus *Neurospora*.

(26) [7] The method according to any one of [1] to [4], wherein a host organism of the transformant is a microorganism of the genus *Aspergillus* selected from the group consisting of *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchuensis*, *Aspergillus usamii*, *Aspergillus aculeatus* and *Aspergillus saitoi*.

(27) [8] A transformant that has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes.

(28) [9] The transformant according to [8], wherein the transformant has two copies to eight copies of the EgtA gene inserted therein and can overexpress the inserted gene.

(29) [10] A transformant that has two copies to eight copies of an EgtA gene inserted therein and can overexpress the inserted gene.

Advantageous Effects of Invention

(30) According to the method or the transformant, which serve as one embodiment of the present invention, selenoneine can be produced at high yields from selenic acid which is an inorganic selenium compound. Therefore, the method or the transformant serving as one embodiment of the present invention allows industrial-scale production of selenoneine.

(31) Furthermore, according to the method or the transformant, which serve as one embodiment of the present invention, selenoneine extracts with a high ratio of selenoneine relative to ergothioneine can be produced. Therefore, the method or the transformant serving as one embodiment of the present invention allows production of high selenoneine content products by simplifying a separation and/or a purification of selenoneine.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A shows the results of measuring the amount of selenoneine in the supernatant fraction obtained from the culture medium of *Aspergillus sojae* transformant prepared in Examples 2 to 5 as described below.

(2) FIG. 1B shows the results of measuring the amount of selenoneine in the fungal cell fraction obtained from the culture medium of *Aspergillus sojae* transformant prepared in Examples 2 to 5 as described below.

(3) FIG. 2 shows the results of measuring the total amount of selenoneine and ergothioneine produced by the *Aspergillus sojae* transformant prepared in Examples 2 to 5 as described below.

(4) FIG. 3 shows the results of calculating the ratio of the amount of selenoneine relative to the total amount of selenoneine and ergothioneine produced by the *Aspergillus sojae* transformant prepared in Examples 2 to 5 as described below.

(5) FIG. 4 shows the results of evaluation of sensitivity of the *Aspergillus sojae* NBRC4239 strain against sodium selenite in Example 7 as described below.

DESCRIPTION OF EMBODIMENTS

(6) A method and a transformant, which provides one embodiment of the present invention, will now be described in details. The technical scope of the present invention is not limited to what is described in this section; rather, the present invention may take various other forms to the extent that its objectives are achieved.

(7) Unless otherwise specified, each term in the specification is used in the sense normally used by those skilled in the art and should not be construed as having an unduly restrictive meaning.

General Description of the Method

(8) The method in one embodiment of the present invention includes the step of applying histidine and a selenium compound to the first transformant to obtain selenoneine, and the first transformant has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes. The first transformant is transformed to overexpress at least one gene selected from the group consisting of the SatA gene, the CysB gene and the MetR gene; that is, any one or two or all three of these genes, in addition to the EgtA gene.

(9) The method in another embodiment of the present invention includes the step of applying histidine and a selenium compound to the second transformant to obtain selenoneine, and the second transformant has two copies to eight copies of an EgtA gene inserted therein and can overexpress the inserted gene. The second transformant is transformed by inserting two copies, three copies, four copies, five copies, six copies, seven copies or eight copies of the EgtA gene.

(10) For example, a transformant with two copies of the EgtA gene inserted refers to a transformant transformed by inserting two DNA constructs prepared in such a manner that the EgtA gene is

appropriately expressed in a host organism into a chromosome. As used herein, a DNA construct prepared so that a specific gene is normally expressed in a host organism is also referred to as an “expression cassette” of the gene. In principle, a transformant with two copies of the EgtA gene inserted expresses the EgtA gene independently from each of two EgtA expression cassettes.

(11) The method in another embodiment of the present invention includes the step of applying histidine and a selenium compound to the third transformant to obtain selenoneine, and the third transformant has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and two copies to eight copies of EgtA gene inserted therein and can overexpress the inserted genes. The third transformant is transformed by inserting all of the genes that are overexpressed by the first transformant and the second transformant.

(12) As used herein, when simply referring to a transformant, “transformant” refers to all of the first transformant, the second transformant and the third transformant. These transformants are another embodiment of the present invention.

(13) As used herein, the selenium compound includes, in addition to selenium compounds themselves, salts, complexes, crosslinked products and derivatives of selenium compounds.

(14) Without wishing to be bound by any theory or presumption, one proposed mechanism of the biosynthesis of selenoneine in fungi can be schematically represented by the Scheme [I] below. The Scheme [I] shows a reaction that produces selenoneine from histidine and selenocysteine via hercynine.

(15) ##STR00001##

wherein PLP represents pyridoxal 5'-phosphate.

(16) EgtA, SatA, CysB and MetR in the Scheme [I] represent proteins encoded by the EgtA gene, the SatA gene, the CysB gene and the MetR gene, respectively.

(17) As shown in the Scheme [I], the EgtA protein catalyzes the reaction in which histidine is converted to hercynine, and the reaction in which hercynine and selenocysteine are converted to hercynylselenocysteine. The SatA protein catalyzes the reaction in which serine is converted to acetylserine. The CysB protein catalyzes the reaction in which acetylserine and hydrogen selenide are converted to selenocysteine. The MetR protein works as a transcription factor, and promotes the reaction in which selenic acid is converted to selenious acid, and further the reaction in which selenious acid is converted to hydrogen selenide. The EgtB and EgtC proteins, which are encoded by the EgtB and EgtC genes, respectively, independently catalyze the reaction in which hercynylselenocysteine is converted to selenoneine.

(18) The transformant for use of the method in another embodiment of the present invention can overexpress the EgtA, SatA, CysB and MetR genes inserted as foreign genes to ultimately produce selenoneine from histidine and a selenium compound.

(19) Copy numbers of the SatA, CysB and MetR genes in the first and third transformants are 1 or 2 or more, respectively. The upper limit on the copy numbers is not particularly limited, but is typically about six.

(20) In addition to the EgtA, SatA, CysB and MetR genes, other foreign genes may be inserted into the transformant. Other foreign genes are not specifically limited as long as they do not prevent the present invention from solving the present problems. Examples include EgtB and EgtC genes. The transformant may be able to efficiently produce selenoneine from hercynylselenocysteine by overexpressing the EgtB, EgtC or both genes inserted as foreign genes. In addition, the insertion of the EgtB or EgtC gene does not significantly change the productivity of selenoneine if the host organism expresses the EgtB or EgtC gene at sufficient levels.

(21) As used herein, genes such as the EgtA, SatA, CysB and MetR genes, which are inserted and overexpressed in transformant may be referred to as inserted genes. Proteins encoded by inserted genes may be collectively referred to as inserted gene proteins.

(22) (Inserted Gene)

(23) An EgtA gene is not limited to any gene corresponding to the “gene encoding enzyme (1)”

described in WO2017/026173 (application number: PCT/JP2016/068128). The EgtA gene can be a gene encoding the EgtA protein; that is an enzyme that has an activity to catalyze the reaction in which hercynylselenocysteine is produced from histidine and selenocysteine in the presence of S-adenosylmethionine and iron (II).

(24) A SatA gene is not limited to any gene encoding a generally known serine O-acetyltransferase (EC2.3.1.30). Serine O-acetyltransferase has an activity to catalyze the reaction in which O-acetyl-L-serine and CoA (coenzyme A) are produced from acetyl-CoA and L-serine.

(25) A CysB gene is not limited to any gene encoding a generally known cysteine synthase (EC2.5.1.47). Cysteine synthase has an activity to catalyze the reaction in which L-cysteine and acetic acid are produced from O-acetyl-L-serine and hydrogen sulfide. Cysteine synthase catalyzes the reaction of O-acetyl-L-serine with sulfide (s). In the present invention, this is applied to the CysB gene to obtain L-selenocysteine from O-acetyl-L-serine and hydrogen selenide (Se), which serves as selenide (Se).

(26) Many microorganisms have been reported to have the SatA and CysB genes. For example, in the document (Microbiology (2000), 146, 2695-2703; GenBank: AAB84208.1; Curr Genet. 1997 Apr. 31(4):348-356), SatA and CysB genes in *Aspergillus nidulans* have been reported. In addition, the document (Research in Microbiology 24 Mar. 2007, 158(5):428-436) reported that many filamentous fungi enzymes have cysteine synthase activity. The assumed conversion of serine and selenic acid to selenocysteine, mediated by proteins encoded by SatA and CysB genes, is shown in the Scheme [II] below.

(27) ##STR00002##

(28) A MetR gene is not specifically limited insofar as it corresponds to “a gene comprising DNA encoding a protein having the function of regulating the expression of sulfur-assimilatory genes” as described in Patent No. JP 4029927. The MetR gene can be a gene that encodes a protein that, as a transcription factor, regulates the expression of a sulfur-assimilatory gene selected from the group consisting of arylsulfatase gene, cholinesulfatase gene, sulfate permease gene and sulfate reductase gene. The expected conversion of sulfuric acid to hydrogen sulfide by proteins encoded by the sulfur-assimilatory genes is shown in the Scheme [III] below.

(29) ##STR00003##

(30) The MetR factor encoded by the MetR gene promotes the reaction in which sulfuric acid is converted to sulfurous acid, and further the reaction in which sulfurous acid is converted to hydrogen sulfide. In the present invention, this is applied to the MetR gene to obtain selenious acid from selenic acid and even hydrogen selenide. The MetR factor regulates the expression of genes encoding enzymes according to sulfur assimilation, particularly inorganic sulfur assimilation. For example, FIG. 9 in the document of Amich et al. (PLOS Pathogens, August 2013 Vol. 9, Issue 8, e1003573) provides an overview of MetR regulation in sulfur metabolism. The function of the MetR factor can prevent the oxidative conversion of selenious acid to selenic acid.

(31) The expected biosynthetic pathway of sulfur assimilation concerning proteins encoded by the SatA, CysB and MetR genes are shown in the Scheme [IV] below. The Scheme [IV] is quoted from FIG. 9 in the above Amich et al.

(32) ##STR00004##

(33) The EgtB and EgtC genes are not limited to any gene corresponding to the “gene encoding enzyme (2)” described in WO 2017/026173. The EgtB and EgtC genes can be genes encoding the EgtB and EgtC proteins; that are enzymes that has an activity to catalyze the reaction in which selenoneine is produced from hercynylselenocysteine using pyridoxal 5'-phosphate (PLP) as a coenzyme.

(34) The inserted genes are overexpressed in the transformant to produce the inserted gene proteins. As used herein, the term “expression of a gene” means that the enzyme encoded by a gene is produced via transcription and translation in a form that exhibits its inherent activities and/or functions. As used herein, the term “overexpression of a gene” means that the protein encoded by

an inserted gene is produced at a level exceeding the normal expression level of the protein in the host organism.

(35) The inserted genes may be a gene that can produce the inserted gene proteins via splicing after the gene inserted into the host organism is transcribed, or alternatively, it may be a gene that can produce the inserted gene proteins without requiring splicing after the transcription of the gene.

(36) The inserted genes may not be completely identical to the inherent gene (i.e., wild-type gene) of the organism of origin: it may be any DNA fragment with a nucleotide sequence that hybridizes to the nucleotide sequence complementary to the nucleotide sequence of the wild-type gene under stringent conditions as long as the gene encodes protein having at least the above-described activities and/or functions.

(37) As used herein, “the nucleotide sequence that hybridizes under stringent conditions” refers to a sequence of DNA obtained by using a hybridization method that uses DNA fragment having the nucleotide sequence of the wild-type gene as a probe. Examples of the hybridization method include colony hybridization, plaque hybridization and southern blot hybridization.

(38) As used herein, the term “stringent condition” refers to a condition under which the signals from specific hybrids can be clearly distinguished from the signals from non-specific hybrids. The stringent condition may vary depending on the hybridization system used, type of the probe, and the sequence and its length. Such conditions may be determined by varying the hybridization temperature or by varying the washing temperature and the salt concentration. For example, if even the signals from non-specific hybrids are strongly detected, the specificity can be increased by increasing the temperature for the hybridization and the washing temperature and if necessary, by decreasing the salt concentration for the washing. In contrast, if even the signals from specific hybrids are not detected, the hybrids may be stabilized by decreasing the temperature for the hybridization and the washing and if necessary, by increasing the salt concentration for the washing.

(39) A specific example of the stringent condition is as follows: using a DNA probe as a probe; the hybridization is carried out overnight (approximately 8 to 16 hours) using 5×SSC, 1.0 (w/v) % blocking reagent for nucleic acid hybridization (Boehringer Mannheim), 0.1 (w/v) % N-lauroylsarcosine, and 0.02 (w/v) % SDS; the washing may be performed twice for 15 min each, using 0.1 to 0.5×SSC and 0.1 (w/v) % SDS, preferably 0.1×SSC and 0.1 (w/v) % SDS. The temperature to carry out the hybridization and the washing is 65° C. or higher, preferably 68° C. or higher.

(40) Examples of the DNA having a nucleotide sequence that hybridizes under stringent conditions include DNA having the nucleotide sequence of the wild-type gene originating from a colony or plaque; DNA obtained by carrying out hybridization under stringent conditions using a filter on which fragments of the DNA are immobilized; and DNA identified by carrying out hybridization at 40 to 75° C. in the presence of 0.5 to 2.0 M NaCl, preferably at 65° C. in the presence of 0.7 to 1.0 M NaCl, and subsequently washing the filter at 65° C. using 0.1 to 1×SSC solution (a 1×SSC solution contains 150 mM sodium chloride and 15 mM sodium citrate). The preparation of the probe and the hybridization can be performed according to the procedures described in textbooks such as *Molecular Cloning: A laboratory Manual*, 2nd-Ed., Cold Spring Harbor Laboratory, Cold Spring Harbor, NY., 1989, *Current Protocols in Molecular Biology*, Supplement 1-38, John Wiley & Sons, 1987-1997 (These literatures will be referred to as reference literature, hereinafter. The entire disclosure of reference literature is incorporated herein by reference). Those skilled in the art would adequately determine the conditions for obtaining DNA having a nucleotide sequence that hybridizes to the nucleotide sequence complementary to the nucleotide sequence of the wild-type gene under stringent conditions by considering, in addition to the above-mentioned conditions such as the salt concentration of buffers and the temperature, other conditions such as the probe concentrations, probe lengths, and the reaction time.

(41) Examples of the DNA having a nucleotide sequence that hybridizes under stringent conditions

include a DNA having a particular percentage or higher sequence identity to the nucleotide sequence of the DNA used as a probe having the nucleotide sequence of the wild-type gene, such as DNA having 80% or higher, preferably 85% or higher, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, or 99% or higher, still more preferably 99.5% or higher sequence identity to the nucleotide sequence of the wild-type gene. The upper limit is not particularly limited, but is typically 100%.

(42) Examples of the nucleotide sequence that hybridizes to a nucleotide sequence complimentary to the nucleotide sequence of the wild-type gene under stringent conditions include nucleotide sequences resulting from deletion, substitution, addition or other modification of from 1 to several, preferably from 1 to 50, more preferably from 1 to 30, even more preferably from 1 to 20, still even more preferably 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 nucleotides, in the nucleotide sequence of the wild-type gene, per unit consisting of 100 nucleotides. As used herein, the term “deletion of a nucleotide” means that a nucleotide is lost or eliminated from the sequence. The term “substitution of a nucleotide” means that a nucleotide is replaced with another nucleotide. The term “addition of a nucleotide” means that a new nucleotide is added to the sequence by inserting it into the sequence.

(43) While the protein-coded by a nucleotide sequence that hybridizes to a nucleotide sequence complementary to the nucleotide sequence of the wild-type gene under stringent conditions should be a protein in having an amino acid sequence resulting from deletion, substitution, addition or other modification of to several amino acids in the amino acid sequence of the enzyme encoded by the nucleotide sequence of the wild-type gene, it has the same activities and/or functions as the protein encoded by the nucleotide sequence of the wild-type gene.

(44) The amino acid sequence of the inserted gene protein may be any amino acid sequence resulting from deletion, substitution, addition or other modification of one to several amino acids in the amino acid sequence of the wild-type enzyme as long as it has the same activities and/or functions as the protein encoded by the inserted gene. As used herein, the range specified by the phrase “one to several” as in “deletion, substitution or addition of one to several amino acids” in the amino acid sequence is not particularly limited but specifically refers to, in the amino acid sequence, per unit consisting of 100 amino acids, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 amino acids, preferably 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 or so amino acids, more preferably 1, 2, 3, 4, or 5 or so amino acids. As used herein, the term “deletion of amino acids” means that amino acid residues are lost or eliminated from the sequence. The term “substitution of amino acids” means that amino acid residues are replaced with other amino acid residues. The term “addition of amino acids” means that new amino acid residues are added to the sequence by inserting them into the sequence.

(45) Specific examples of “deletion, substitution or addition of one to several amino acids” include embodiments in which one to several amino acids are replaced with other chemically similar amino acids. For example, a hydrophobic amino acid may be substituted with another hydrophobic amino acid, or a polar amino acid may be substituted with another polar amino acid having the same charge. Such chemically similar amino acids are known in the art for each amino acid. Specific examples of non-polar (hydrophobic) amino acids include alanine, valine, isoleucine, leucine, proline, tryptophan, phenylalanine, and methionine. Examples of polar (neutral) amino acids include glycine, serine, threonine, tyrosine, glutamine, asparagine and cysteine. Examples of positively charged basic amino acids include arginine, histidine and lysine. Examples of negatively charged acidic amino acids include aspartic acid and glutamic acid.

(46) Examples of the amino acid sequences resulting from deletion, substitution, addition or other modification of one to several amino acids in the amino acid sequence of the wild-type enzyme include amino acid sequences having a particular percentage or higher sequence identity to the amino acid sequence of the wild-type enzyme, such as amino acid sequences having 80% or higher,

preferably 85% or higher, more preferably 90% or higher, 91% or higher, 92% or higher, 93% or higher, 94% or higher, 95% or higher, 96% or higher, 97% or higher, 98% or higher, or 99% or higher, still more preferably 99.5% or higher sequence identity to the amino acid sequence of the wild-type enzyme.

(47) (Means for Determining Sequence Identity)

(48) While the sequence identity between nucleotide sequences or amino acid sequences may be determined by any method, it can be determined by using a commonly known method, whereby a wild-type gene or an amino acid sequence of an enzyme encoded by the wild-type gene is aligned with a nucleotide sequence or amino acid sequence of interest and the percent match between the two sequences is calculated using a program.

(49) The algorithm of Karlin and Altschul is a known program for determining the percent match between two amino acid sequences or nucleotide sequences (Proc. Natl. Acad. Sci. USA 87: 2264-2268, 1990; Proc. Natl. Acad. Sci. USA 90: 5873-5877, 1993). Using this algorithm, Altschul et al. developed the BLAST program (J. Mol. Biol. 215: 403-410, 1990). The Gapped BLAST program, which can determine the sequence identity in a more sensitive way than the BLAST, is also known (Nucleic Acids Res. 25: 3389-3402, 1997). Using the above-described programs, one skilled in the art can search in a database for a sequence with a high sequence identity to a given sequence. These programs are available on the website of U.S. National Center for Biotechnology Information (blast.ncbi.nlm.nih.gov/Blast.cgi).

(50) While the above-described methods are commonly used in the search of sequences with certain sequence identities from a database, Genetyx network model, version 12.0.1 (Genetyx corporation) may also be used in a homology analysis to determine the sequence identity of individual sequences. This method is based on the Lipman-Pearson method (Science 227:1435-1441, 1985). When analyzing the sequence identity of nucleotide sequences, regions encoding proteins (CDS or ORF) are used when possible.

(51) (Origins of Inserted Genes)

(52) The inserted genes are, for example, derived from species having the ability to produce selenoneine or the ability to produce ergothioneine, or species expressing the inserted genes. Examples of the organisms of origin from which the inserted genes are derived include microorganisms. Of various microorganisms, filamentous fungi are preferred since many of their species are known to have the ability to produce ergothioneine. Examples of the filamentous fungi include fungi of the genus *Aspergillus*. Specific examples include *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchensis*, *Aspergillus usamii*, *Aspergillus aculeatus*, *Aspergillus saitoi* and *Aspergillus nidulans*.

(53) *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchensis*, *Aspergillus usamii*, *Aspergillus aculeatus* and *Aspergillus saitoi* listed above as specific examples of the filamentous fungi of the genus *Aspergillus* have long been used in the production of miso paste, soy sauce, Japanese sake, shochu liquor and other fermented products, as well as in the production of citric acid and enzymes such as amylases. Their high enzyme productivity and high reliability for the safety, backed by a long history of use, make these microorganisms highly useful in industrial applications.

(54) As described above, while the organisms of origin from which the inserted genes are derived are not particularly limited, the inserted: gene proteins expressed in the transformant might not be deactivated by the growth conditions of the host organisms or the proteins might show their respective activities or functions. For this reason, it is preferred that the organism of origin from which the inserted genes are derived be a microorganism that grows under conditions similar to the growth conditions of a host organism to be transformed by the insertion of the inserted genes.

(55) (Cloning of Inserted Genes Using Genetic Engineering Technique)

(56) The inserted genes can be inserted into various suitable known vectors. The resulting vector can then be introduced into a suitable known host organism to create a transformant in which the

recombinant vector (recombinant DNA) containing the inserted genes has been introduced. A person skilled in the art can appropriately select a suitable method for obtaining the inserted genes, a method for obtaining the inserted gene sequence information and the amino acid sequence information of the inserted gene protein, as well as a method for creating different vectors and a method for creating transformants. The terms “transformation” and “transformant” as used herein encompass transduction and transductants, respectively. One non-limiting example of cloning of the inserted genes will be described below.

(57) Cloning of the inserted genes may suitably use commonly used gene cloning techniques. For example, using a standard technique such as the technique described in the reference literature, the chromosomal DNA and mRNA can be extracted from microorganisms and various cells capable of producing the inserted gene proteins. The extracted mRNA can be used as a template to synthesize cDNA. The resulting chromosomal DNA and cDNA may be used to construct a library of chromosomal DNA or cDNA.

(58) For example, inserted genes can be obtained by cloning from the chromosomal DNA or cDNA derived from microorganisms having the genes, which serves as a template. The organisms of origin from which the inserted genes are derived are as described above; specific examples include *Aspergillus sojae* NBRC4239 strain and *Aspergillus oryzae* RIB40 strain. For example, the *Aspergillus sojae* NBRC4239 strain is cultured and the resulting fungal cells are dehydrated and physically triturated using a mortar while chilled in liquid nitrogen to form fine powder-like cells debris, from which a fraction containing chromosomal DNA is extracted using a standard technique. A commercially available DNA extraction kit such as DNeasy Plant Mini Kit (Qiagen) can be used to extract the chromosomal DNA.

(59) Subsequently, a polymerase chain reaction (referred to as PCR, hereinafter) was conducted using the chromosomal DNA as a template along with synthetic primers complementary to the sequences at the 5' and 3' ends. The primers are not particularly limited as long as they can amplify DNA fragments containing the inserted gene. Examples of the primers include primers shown in SEQ ID NOs: 19 to 20, 45 to 46, 54 to 55 and 58 to 59 designed based on the genome sequence of *Aspergillus sojae*. These primers can amplify the full length of the target gene and can therefore eliminate the need for RACE. Alternatively, DNA sequences containing fragments of the target gene may be amplified using suitable PCR techniques such as 5' RACE and 3' RACE and these sequences are subsequently ligated to obtain a DNA segment containing the full length target gene.

(60) The method for obtaining the inserted gene is not particularly limited; for example, rather than using genetic engineering techniques, the inserted gene may be constructed by chemical synthesis.

(61) For example, the nucleotide sequences of the amplification products amplified by PCR and the chemically synthesized genes may be determined as follows. First, the DNA segment to be sequenced is inserted into a suitable vector according to the standard technique to prepare a recombinant DNA. For cloning into a vector, a commercially available kit, such as TA Cloning Kit (Invitrogen); commercially available plasmid vector DNA, such as pUC119 (Takara Bio), pUC18 (Takara Bio), pBR322 (Takara Bio), pBluescript SK+(Stratagene), and pYES2/CT (Invitrogen); and commercially available bacteriophage vector DNA, such as λEMBL3 (Stratagene), may be used. The recombinant DNA is then used to transform host organisms, such as *Escherichia coli*, preferably *E. coli* JM109 strain (Takara Bio) and *E. coli* DH5a strain (Takara Bio). The recombinant DNA present in the transformant is then purified using a purification kit such as QIAGEN Plasmid Mini Kit (Qiagen).

(62) The nucleotide sequences of genes inserted in the recombinant DNA are then determined by the dideoxy sequencing technique (Methods in Enzymology, 101, 20-78, 1983). The sequence analyzer used to determine the nucleotide sequence is not particularly limited; for example, Li-COR MODEL 4200L sequencer (Aloka), 370DNA sequencing system (Perkin Elmer), CEQ2000XL DNA analysis system (Beckman) may be used. The determined nucleotide sequences may then be used to estimate the amino acid sequence of the translated proteins, thus, the inserted

gene proteins.

(63) (Construction of a Recombinant Vector Containing Inserted Genes)

(64) Recombinant vectors containing the inserted genes (recombinant DNA) can be constructed by connecting a PCR amplification product containing any of the inserted genes with any of various vectors in such a manner that the recombinant vector can express the inserted genes. For example, such a recombinant vector may be constructed by excising a DNA fragment containing any of the inserted genes with appropriate restriction enzyme and ligating the DNA fragment into a plasmid cut with appropriate restriction enzyme. The recombinant vector may also be obtained by connecting a DNA fragment containing the gene and having sequences homologous to a plasmid attached to the both ends with a DNA fragment derived from the plasmid amplified by inverse PCR using a commercially available recombinant vector preparation kit such as In-Fusion HD Cloning Kit (Clontech).

(65) (Method for Creating a Transformant)

(66) The method for creating a transformant is not particularly limited. For example, a transformant can be created by inserting a gene into a host organism in which the inserted genes are expressed according to standard method. Specifically, a DNA construct in which any of the inserted genes has been inserted between an expression-inducing promoter and a terminator is constructed.

Subsequently, a host organism is transformed with the DNA construct containing the inserted genes to obtain a transformant that overexpresses the inserted genes. In the present specification, DNA fragments comprising an expression-inducing promoter—inserted genes—a terminator and recombinant vectors containing the DNA fragment that are prepared to transform the host organism are collectively referred to as “DNA constructs” or “inserted gene expression cassettes.”

(67) The method for inserting the inserted genes in a host organism in such a manner that the inserted genes are expressed in the host organism is not particularly limited. For example, the gene may be directly inserted into the chromosome of the host organism by making use of homologous recombination or non-homologous recombination, or the gene may be connected to a plasmid vector, which in turn is introduced into the host organism.

(68) In the method that makes use of homologous recombination, a DNA construct may be connected between sequences homologous to the upstream region and the downstream region of a recombination site on a chromosome and inserted into the genome of the host organism. As a result of this self-cloning, a transformant can be obtained in which the gene is overexpressed under control of a high expression promoter in the DNA construct. The high expression promoter may be any high expression promoter, including, for example, a promoter region of translation elongation factor TEF1 gene (*tef1*), a promoter region of α -amylase gene (*amy*), a promoter region of alkaline protease gene (*alp*), and other suitable promoters.

(69) In the method that makes use of a vector, a DNA construct is integrated into a plasmid vector used to transform host microorganisms using a standard method and a corresponding host organism can be transformed with the plasmid vector according to a standard method.

(70) A suitable vector—host system may be any system that allows the production of the inserted gene proteins in the host organisms, including, for example, a system based on pUC19 and a filamentous fungus, and a system based on pSTA14 (Mol. Gen. Genet. 218, 99-104, 1989) and a filamentous fungus.

(71) While the DNA construct is preferably introduced into the chromosome of the host organisms, it may be used without introducing into the chromosome by integrating into a self-replicating vector (Ozeki et al. Biosci. Biotechnol. Biochem. 59, 1133 (1995)).

(72) The DNA construct may contain a marker gene that allows the selection of transformed cells. Examples of the marker gene include, but are not limited to, genes compensating for the nutritional requirements of the host organisms, such as *pyrG*, *niaD* and *adeA*; and drug-resistant genes such as those against pyrithiamine, hygromycin B and oligomycin. Also, the DNA construct preferably contains a promoter, a terminator and other regulatory sequences (such as enhancer and

polyadenylated sequences) that enable the overexpression of the inserted genes in the host organisms. The promoter may be any suitable expression-inducing promoter or constitutive promoter, including, for example, tef1 promoter, alp promoter, and amy promoter. The terminator may also be any terminator, including, for example, alp terminator, amy terminator, and tef1 terminator.

(73) The regulatory sequences for the inserted genes in the DNA construct are not necessarily required if the DNA fragments containing the inserted genes contain sequences having expression regulatory functions. Also, when transformation is performed by the cotransformation method, the DNA construct may not contain any marker genes.

(74) Purification tags of the inserted gene proteins may be added to the DNA construct. For example, a suitable linker sequence may be added to the upstream or downstream of the inserted genes and six or more codons of histidine-encoding nucleotide sequences may be added to the linker to enable the purification on a nickel column.

(75) One example of the DNA construct is a DNA construct in which a tef1 gene promoter, an inserted gene, an alp gene terminator and a pyrG marker gene are connected to the In-Fusion cloning Site located in the multiple cloning site of pUC19.

(76) Any properly selected method known to those skilled in the art may be used for transformation into filamentous fungi; for example, the protoplast PEG technique in which protoplasts of a host organism are prepared and polyethylene glycol and calcium chloride are added may be used (See, for example, Mol. Gen. Genet. 218, 99-104, 1989, Japanese Unexamined Patent Application Publication No. 2007-222055). The medium to regenerate the transformant is properly selected depending on the host organism and the transformation marker gene used. For example, when *Aspergillus sojae* is used as the host organism and pyrG gene is used as the transformation marker gene, the transformant can be regenerated in a Czapek-Dox minimal medium (Difco) containing 0.5% agar and 1.2M sorbitol.

(77) Alternatively, in order to obtain the transformant, the endogenous promoter for the inserted genes present on the chromosome of the host organism may be substituted with a high expression promoter such as tef1 by homologous recombination. Again, a transformation marker gene such as pyrG is preferably inserted in addition to the high expression promoter. For example, a transformation cassette consisting of the upstream region of the inserted genes—a transformation marker gene—a high expression promoter—all or a part of the inserted genes may be used for this purpose (see Example 1 and FIG. 1 of Japanese Unexamined Patent Application Publication No. 2011-239681). In this case, the upstream region of the inserted genes and all or a part of the inserted genes are used in homologous recombination. The all or a part of the inserted genes used may include a region of the gene extending from the start codon to somewhere down the length of the gene. A suitable length of the region is preferably 0.5 kb or longer for homologous recombination.

(78) The creation of the transformant can be confirmed by the following procedure: the transformant may be cultured under a condition that induces the activities or functions of the inserted gene proteins and subsequently the resulting culture may be examined for the presence of selenoneine; or alternatively, a comparison may be made to determine if the amount of selenoneine present in the resulting culture is greater than the amount of selenoneine present in a culture of the host organism cultured under the same condition.

(79) The creation of the transformant can be also confirmed by the following procedure: chromosomal DNA is extracted from the transformant and PCR is performed using the chromosomal DNA as a template to detect the presence of any PCR product that can be amplified if the transformation has occurred.

(80) For example, a PCR can be performed using a combination of a forward primer for the nucleotide sequence of the promoter used and a reverse primer for the nucleotide sequence of the transformation marker gene and whether the product having an expected length is produced is

determined.

(81) When the transformation is carried out by homologous recombination, it is preferred to perform a PCR using a forward primer located upstream of the upstream homologous region used and a reverse primer located downstream of the downstream homologous region used and then determine whether the product having a length expected when the homologous recombination has occurred is produced.

(82) By inserting a second inserted gene into the transformant in which the first inserted gene is inserted, a transformant in which the first inserted gene and the second inserted gene are inserted can be prepared. By repeating this procedure, a transformant with multiple inserted genes can be prepared.

(83) When preparing a transformant in which multiple inserted genes are inserted, if the marker gene inserted together with the first inserted gene and the marker gene inserted together with the second inserted gene are the same, it is preferable to remove the marker gene inserted together with the first inserted gene after the first inserted gene is inserted.

(84) For example, in an inserted gene expression cassette containing a first inserted gene, a region for looping out and a marker gene which are linked in order, if a region with the same sequence as the region for looping out is present in downstream of a site on the host organism's chromosome that can be introduced by homologous recombination, the inserted gene expression cassette is inserted onto the host organism's chromosome due to homologous recombination that occurs between the region for looping out and the same sequence region downstream so that the marker gene can be removed by looping out. After the first inserted gene has been inserted, a transformant can be obtained in which the inserted marker gene along with the first inserted gene is removed.

(85) (Host Organism)

(86) The host organism is not limited to any microorganism that can produce the inserted gene proteins when transformed by a DNA construct containing the inserted genes. An example thereof is a microorganism capable of metabolizing selenium in view of the toxicity of the selenium compound. The host organism is preferably microorganisms that can express selenic acid reductase (EC1.97.1.9), selenocysteine lyase (EC4.4.1.16), serine dehydratase (EC4.3.1.17) or two or more of those enzymes, more preferably filamentous fungi such as the genus *Aspergillus*, the genus *Escherichia*, the genus *Trichoderma*, the genus *Fusarium*, the genus *Penicillium*, the genus *Rhizopus* and the genus *Neurospora*, photosynthetic microorganism and probiotic microorganism.

(87) For example, it is known that microorganisms such as the genus *Acinetobacter*, the genus *Aeromonas*, the genus *Arthrobacter*, the genus *Bacillus*, the genus *Candida*, the genus *Cephalosporium*, the genus *Citrobacter*, the genus *Corynebacterium*, the genus *Flavobacterium*, the genus *Fusarium*, the genus *Micrococcus*, the genus *Neurospora*, the genus *Penicillium*, the genus *Pseudomonas*, the genus *Salmonella*, the genus *Scopulariopsis*, the genus *Selenomonas* have an oxidation or reducing ability for selenium compound (refer to D. T. Maier et al., APPLIED AND ENVIRONMENTAL MICROBIOLOGY, October 1988, p.2591-2593). Especially, selenate reductase or the gene encoding the enzyme is found from *Thauera selenatis*, *Escherichia coli*, *Enterobacter cloacae* and *Bacillus selenatarsenatis* (refer to SAKAGUCHI Toshifumi, "selenium oxyanion reductase and its gene", Biomedica, 2012 Vol. 3, p.133). Also, it is known that *Alcaligenes viscolactis*, *Escherichia freundii*, *Corynebacterium pseudodiphtheriticum*, *Pseudomonas alkanolytica*, *Brevibacterium leucinophagum*, *Escherichia coli*, *Erwinia carotovora*, *Serratia marcescens*, *Alcaligenes bookeri*, *Aspergillus ficuum*, *Aspergillus sojae*, *Absidia corymbifera*, *Neurospora crassa*, *Penicillium expansum*, *Saccharomyces cerevisiae*, *Kluyveromyces fragilis*, *Candida albicans*, *Hansenula beckii* and *Schwanniomyces occidentalis* have a selenocysteine lyase activity or a possibility of said activity (refer to PATRICK CHOCAT et al., JOURNAL OF BACTERIOLOGY, October 1983, p.455-457). Thus, those microorganisms can be used as host organisms. Also, beyond those microorganisms, any other microorganisms having the reinforced selenium metabolism gene or the expression of the heterologous gene can be used as host

organisms. Further, it may be possible that the microorganism can be used as the organism of origin from which the inserted genes are derived.

(88) Among them, the host organism is more preferably any of the microorganisms of filamentous fungi in which the production of ergothioneine is detected and filamentous fungi that have the inserted genes on their genomic DNA. Specific examples of the filamentous fungi include filamentous fungi described in Donald et al. document (Donald B. Melville et al., J. Biol. Chem. 1956, 223:9-17, the entire disclosure of which is incorporated herein by reference) and Dorothy et al. document (Dorothy S. Genghof, J. Bacteriology, August 1970, p.475-478, the entire disclosure of which is incorporated herein by reference), such as filamentous fungi belonging to the genus *Aspergillus*, the genus *Neurospora*, the genus *Penicillium*, the genus *Fusarium*, the genus *Trichoderma*, and the genus *Mucor*. Examples of the filamentous fungi that have inserted genes on their genomic DNA include filamentous fungi belonging to the genus *Neosartorya*, the genus *Byssochlamys*, the genus *Talaromyces*, the genus *Ajellomyces*, the genus *Paracoccidioides*, the genus *Uncinocarpus*, the genus *Coccidioides*, the genus *Arthroderma*, the genus *Trichophyton*, the genus *Exophiala*, the genus *Capronia*, the genus *Cladophialophora*, the genus *Macrophomina*, the genus *Leptosphaeria*, the genus *Bipolaris*, the genus *Dothistroma*, the genus *Pyrenophora*, the genus *Neofusicoccum*, the genus *Setosphaeria*, the genus *Baudoinia*, the genus *Gaeumannomyces*, the genus *Marssonina*, the genus *Sphaerulina*, the genus *Sclerotinia*, the genus *Magnaporthe*, the genus *Verticillium*, the genus *Pseudocercospora*, the genus *Colletotrichum*, the genus *Ophiostoma*, the genus *Metarhizium*, the genus *Sporothrix* and the genus *Sordaria*.

(89) Of these filamentous fungi, in terms of the safety and easy culturing, the host filamentous fungus is preferably any of the microorganisms of the genus *Aspergillus* listed above as the organisms of origin from which the inserted genes are derived, including *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchensis*, *Aspergillus usamii*, *Aspergillus aculeatus*, *Aspergillus saitoi* and *Aspergillus nidulans*.

(90) The host organism may be a wild strain or a transformant obtained by transforming the wild strain in advance. The transformant obtained by transforming a wild strain in advance to be used as a host organism is not particularly limited.

(91) Filamentous fungi, including the microorganisms of the genus *Aspergillus* tend to have a low frequency of homologous recombination. Therefore, in case that the transformant is prepared by homologous recombination, it is preferable to use transformed filamentous fungi in which Ku genes such as Ku70 and Ku80 involved in the non-homologous recombination mechanism is suppressed.

(92) Such suppression of the Ku gene can be carried out by any method known to those skilled in the art. For example, the Ku gene can be suppressed by disrupting the Ku gene using a Ku gene disruption vector, or by inactivating the Ku gene by an antisense RNA method using an antisense expression vector of the Ku gene. The homologous recombination frequency of the transformed microorganisms of the genus *Aspergillus* obtained is markedly increased compared to the homologous recombination frequency of the original microorganisms of the genus *Aspergillus* prior to the genetic manipulation of the Ku gene for suppression, specifically at least 2-fold, preferably at least 5-fold, preferably at least 10-fold, preferably at least about 50-fold increased.

(93) As a host organism, it is preferable to use a transformant in which a marker gene such as pyrG is suppressed. The marker gene to be suppressed can be appropriately selected according to the marker gene to be included in the DNA construct.

(94) (Specific Examples of Inserted Genes)

(95) Examples of the inserted genes derived from the *Aspergillus sojae* NBRC4239 strain include AsEgtA, AsSatA, AsCysB and AsMetR genes which will be described in Examples below. The nucleotide sequences of the AsEgtA, AsSatA, AsCysB and AsMetR genes are shown in SEQ ID NOs: 1 to 4 in the sequence listing, respectively. Further, the amino acid sequences of the AsEgtA, AsSatA, AsCysB and AsMetR proteins are shown in SEQ ID NOs: 5 to 8 in the sequence listing,

respectively.

(96) Inserted genes may be obtained from microorganisms other than those of *Aspergillus sojae* by any suitable method. For example, a homology search by BLAST may be conducted on the genomic DNA of microorganisms other than those of *Aspergillus sojae* based on the nucleotide sequences of the AsEgtA, AsSatA, AsCysB and AsMetR genes (SEQ ID NOs: 1 to 4) and the amino acid sequences of the AsEgtA, AsSatA, AsCysB and AsMetR proteins (SEQ ID NOs: 5 to 8), to identify genes having a nucleotide sequence with a high sequence identity to the nucleotide sequences of the AsEgtA, AsSatA, AsCysB and AsMetR genes. Alternatively, inserted genes may be obtained by identifying proteins having a high sequence identity to the AsEgtA, AsSatA, AsCysB and AsMetR proteins from the total protein of microorganisms other than those of *Aspergillus sojae* and identifying the genes encoding these proteins. Whether the resulting genes are equivalent to the inserted genes can be determined by transforming the organism of origin as the host organism with the obtained gene and determining if selenoneine is produced or determining if the production of selenoneine is enhanced compared to the host organisms.

(97) For example, the EgtA gene derived from *Aspergillus oryzae* RIB40 strain include the AoEgtA gene of SEQ ID NO: 23 in the sequence listing in WO2017/026173. The amino acid sequence of the AoEgtA protein is listed in WO2017/026173 as SEQ ID NO: 24 in the sequence listing.

(98) Since *Aspergillus sojae*, *Aspergillus oryzae* and *Aspergillus niger* grow under similar conditions, it may be possible to transform these microorganisms each other by inserting the gene possessed by one of these microorganisms into one of these microorganisms. For example, an inserted gene derived from *Aspergillus sojae* may be inserted into the host organism of *Aspergillus oryzae* or *Aspergillus niger* to transform them. In order to ensure that the inserted gene proteins have the desired enzymatic activity, it is preferred that the organism of origin from which the inserted genes are derived and the host organism are identical. For example, an inserted gene derived from *Aspergillus sojae* may be inserted into the same *Aspergillus sojae*.

(99) The inserted genes may be genes optimized for their codons, secondary structures and GC contents based on the amino acid sequence of the inserted gene proteins derived from *Aspergillus sojae*. Specific examples of such genes include EcEgtA and EcEgtC synthesized for expression in *E. coli*, which are described as SEQ ID NOs: 27 and 28 in the sequence listing in WO2017/026173.

(100) (One embodiment of transformant) One embodiment of the transformant is an *Aspergillus sojae* transformant obtained by inserting two copies to four copies of the AsEgtA gene into *Aspergillus sojae* for overexpression of the AsEgtA protein. Another embodiment of the transformant is an *Aspergillus sojae* transformant obtained by inserting one copy to four copies of the AsEgtA gene, one copy to two copies of the AsSatA gene and one copy to two copies of the AsCysB gene into *Aspergillus sojae* for overexpression of the AsEgtA, AsSatA and AsCysB proteins. Another embodiment of the transformant is an *Aspergillus sojae* transformant obtained by inserting one copy to four copies of the AsEgtA gene and one copy to two copies of the AsMetR gene into *Aspergillus sojae* for overexpression of the AsEgtA and AsMetR proteins. Another embodiment of the transformant is an *Aspergillus sojae* transformant obtained by inserting one copy to four copies of the AsEgtA gene, one copy to two copies of the AsSatA gene, one copy to two copies of the AsCysB gene and one copy to two copies of the AsMetR gene into *Aspergillus sojae* for overexpression of the AsEgtA, AsSatA, AsCysB and AsMetR proteins. Another embodiment of the transformant is an *Aspergillus oryzae* transformant obtained by inserting two copies to four copies of the AsEgtA gene into *Aspergillus oryzae* for overexpression of the AsEgtA protein.

(101) The above-mentioned *Aspergillus sojae* and *Aspergillus oryzae* transformants are designed to overexpress the inserted gene proteins and are capable of produce selenoneine at detectable or higher levels while the respective host organisms can produce little or no selenoneine. In addition, the *Aspergillus sojae* and *Aspergillus oryzae* transformants can produce selenoneine not only from organic selenium compounds such as selenocysteine and selenocystine, but also from inorganic

selenium compounds such as selenious acid, as will be described later in Examples. Accordingly, the transformant is preferably a transformant in which the expression of the inserted genes is enhanced such that the amount of selenoneine is increased as compared to the host organism.

(102) As will be described later in Examples, the *Aspergillus sojae* transformant transformed to overexpress the inserted genes is capable to produce selenoneine. Specifically, the *Aspergillus sojae* transformant can produce selenoneine by culturing in DPY medium suitable for the growth of the host *Aspergillus sojae* at 30° C. for 1 to 3 days to promote fungal cell growth, and then adding sodium selenite to this culture system and culturing at 37° C. for 3 to 5 days. The produced selenoneine is secreted outside the fungal cells or accumulated in the culture medium by lysis of fungal cells.

(103) The ability to produce selenoneine of the transformant is not particularly limited. For example, when the transformant is cultured in DPY medium at 30° C. for 2 days, and then sodium selenite is added to this culture system and further cultured at 37° C. for 4 days, the production of selenoneine is 5 mg/l or more, preferably 10 mg/l or more, more preferably 15 mg/l or more, even more preferably 17 mg/l or more, and even more preferably 20 mg/l or more. The upper limit of selenoneine production when cultured in this manner is not limited and is typically around 100 mg/l.

(104) As will be described later in Examples, the *Aspergillus sojae* transformant transformed to overexpress the inserted genes tends to have a greater ratio of selenoneine against the total amount of selenoneine and ergothioneine produced. Regarding the ability to produce selenoneine of the transformant, for example, when the transformant was incubated in DPY medium at 30° C. for 2 days, and when sodium selenite was added to this culture system and further incubated at 37° C. for 4 days, the ratio of the amount of selenoneine against the total amount of selenoneine and ergothioneine produced is 5% or more, more preferably 10% or more, even more preferably 20% or more, and still more preferably 40% or more. The upper limit of the above ratio when cultured in this manner is not particularly limited and is typically 100%. The upper limit of the above ratio when cultured in this manner is not particularly limited and is typically 100%.

(105) The transformant may produce, along with the proteins produced by the inserted gene, wild-type proteins that have the same or different structural properties from the inserted gene proteins depending on the genes originally possessed by the host organism. Consequently, even if the transformant is not transformed with the inserted gene, the transformant may still be able to produce selenoneine.

(106) (Method)

(107) One embodiment of the method is a method for producing selenoneine including the step of applying histidine and a selenium compound to a transformant that has the inserted genes inserted therein and that can overexpress the inserted genes, to obtain selenoneine.

(108) The method for applying histidine and a selenium compound to the transformant is not particularly limited and may be any method that can expose the transformant to histidine and the selenium compound to allow the activities or functions of inserted gene proteins to produce selenoneine. For example, the transformant may be cultured in a medium containing histidine and selenium compound and optimized for the growth of the transformant under culture conditions suitable for the transformant so as to produce selenoneine. The culture method is not particularly limited; for example, the solid culture or liquid culture technique performed under aerated or non-aerated condition may be employed. The amount of the selenium compound added is not particularly limited as long as the growth of the transformant is not inhibited. For example, the selenium compound may be present at sufficiently low levels relative to the fungal cell concentration at the initial stage of culturing. Specifically, it is added at a concentration of 0 mM. When it is desired to obtain large amounts of selenoneine, the amount of the selenium compound added may be increased during the course of culture or as the fungal cell concentration increases. For example, additional amounts of the selenium compound at a concentration of 0.01 to 10 mM,

preferably 0.1 to 5 mM, may be added to the culture medium 1 to 3 days after the start of culture.

(109) The medium may be any standard medium designed for culturing host organism and may be either a synthetic or natural medium that contains a carbon source, a nitrogen source, inorganic materials, and other nutrients at an appropriate ratio. When the host organism is a microorganism of the genus *Aspergillus*, the DPY medium as described in Examples below may be used, although not particularly limited. It is preferred, however, that the medium contain, as a component, iron (II) required for the activation of the EgtA protein. While iron (II) may be added to the medium in the form of a compound, it may also be added as a mineral-containing material.

(110) The selenium compound is not particularly limited as long as it contains selenium as a constituent element. For example, it may be an organic or inorganic selenium compound or a salt thereof. Examples of organic selenium compounds and salts thereof include selenocysteine, selenocystine, selenomethionine, Se-(methyl)seleno-L-cysteine, selenopeptides, selenoproteins and salts thereof and selenium yeast. Examples of inorganic selenium compounds and salts thereof include selenic acid, selenious acid, selenium chloride, selenium, selenides, selenium sulfide, dimethylselenium, selenophosphate, selenium dioxide and salts thereof. Alternatively, the selenium compound may be an organic material containing an organic or inorganic selenium compound or a salt thereof. Examples of such organic materials include, but are not limited to, bonito fish (processed products and dried bonito), mustard (powdered mustard, grain mustard and mustard paste), pork (kidney, liver, and raw meat), beef (kidney, raw meat), anglerfish (liver, raw meat), codfish (cod roe, raw meat), bluefin tuna (red meat, raw meat), flatfish (raw meat), bonito fish (those caught in the fall season, raw meat), snow crabs (raw meat), sunflower seeds (fried, flavored), horse mackerel (grilled), tilefish (raw meat), granular seasoning, yellow fin tuna (raw meat), albacore (raw meat), oyster (boiled), and other food products known to be a rich source of selenium. The selenium compound may be one of or a combination of two or more of these materials.

(111) The transformant in which the inserted gene is inserted and which overexpress the inserted gene is suitable for producing selenoneine from selenic acid and selenate such as sodium selenite and potassium selenite. Selenic acid and selenate are more economical than organic selenium compounds such as selenocysteine and selenocystine. In addition, selenic acid and selenate are less cell sensitive than selenious acid and may have less adverse effects on selenoneine production. Therefore, selenic acid and selenate are preferred as the selenium compounds to be used.

(112) The culture condition of the transformant may be any culture condition of the host organism commonly known to those skilled in the art. For example, when the host organism is a filamentous fungus, the initial pH of the medium may be conditioned to 5 to 10 and the culture temperature to 20 to 40° C., and the culture time may be properly selected and may vary from several hours to several days, preferably from 1 to 7 days, and more preferably from 2 to 4 days. The culture means is not particularly limited; for example, an aerated and agitated deep culture, a shake culture, a static culture or other suitable culture techniques may be employed with the culture condition preferably adjusted so that sufficient amounts of dissolved oxygen are present. One example of the culture medium and culture condition for culturing microorganisms of the genus *Aspergillus* is as follow: a DPY medium, described in the Examples below, is cultured by agitation or shaking at 30° C., 10-300 rpm for 1 to 3 days for the purpose of fungal cell growth and by agitation or shaking at 37° C., 10-300 rpm for 3 to 5 days for the purpose of selenoneine production.

(113) The method for extracting selenoneine from the culture after completion of the culture is not particularly limited. The culture supernatant recovered from the culture by filtration, centrifugation, or the like may be used for extraction as is, or the culture supernatant concentrate, in which the recovered culture supernatant is dried and/or concentrated, may be used for extraction.

(114) The solvent used for extraction may be any solvent that can dissolve selenoneine, including, for example, organic solvents, such as methanol, ethanol, isopropanol and acetone; water-containing organic solvents composed of these organic solvents and water mixed together; and

water, warm water and hot water. After addition of the solvent into the culture supernatant or the culture supernatant concentrate, selenoneine is extracted while the fungal cells are triturated as necessary. The temperature of the extraction solvent may be set to from room temperature to 100° C.

(115) In order to purify selenoneine, the resulting extract can be subjected to various purification processes including centrifugation, filtration, ultrafiltration, gel filtration, separation by solubility difference, solvent extraction, chromatography (adsorption chromatography, hydrophobic interaction chromatography, cation exchange chromatography, anion exchange chromatography, and reversed-phase chromatography), crystallization, active carbon treatment, membrane treatment, and other purification processes.

(116) The qualitative or quantitative analysis technique of selenoneine may be conducted by, for example, LC-ICP-MS, LC-MS/MS, LC-MS or HPLC according to the condition described in WO2017/026173 or Yamashita et al. (THE JOURNAL OF BIOLOGICAL CHEMISTRY VOL. 285, NO. 24, pp. 18134-18138, Jun. 11, 2010, EXPERIMENTAL PROCEDURES”, “Selenium Determination”). A person skilled in the art would properly select the conditions for the analysis; for example, the analysis may be performed using the conditions described in Examples below.

(117) According to the method in one embodiment of the present invention, selenoneine can be obtained at high yields. In the method described in WO2017/026173 of Patent Document 2, selenoneine is obtained from the cultured cells. In comparison, in the method in one embodiment of the present invention, selenoneine can be obtained from the culture supernatant. In the method in one embodiment of the present invention, selenoneine may be obtained by extraction from the cultured cells properly according to the method described in WO2017/026173 of Patent Document 2.

(118) In the method in one embodiment of the present invention, various other steps or manipulations may be performed before, after, or during the above-described step as long as the objectives of the present invention can be achieved.

(119) (Applications of Selenoneine)

(120) Having advantageous characteristics of being a functional biological material with various physiological activities, as well as being a heat-resistant, water-soluble material, the selenoneine obtained by the method or the transformant to serve as one embodiment of the present invention is useful as general food and beverage products, functional food and beverage products, food and beverage products with function claims, food and beverage products for specified health use, food and beverage products with nutrient function claims, food and beverage products with health function claims, food and beverage products for special uses, nutritional supplement food and beverage products, health-promoting food and beverage products, supplements, beauty food and beverage products, cosmetic products, pharmaceutical products, quasi-pharmaceutical products, animal feeds, and raw-materials for producing these products.

(121) Specifically, selenoneine is known to have antioxidant activity that is 1,000 times as high as that of its thio analog, ergothioneine. For this reason, selenoneine can be useful as a biological antioxidant that exhibits the ability to capture hydroxyl radicals, the ability to suppress autoxidation of the hem iron, and other antioxidant activities. Examples of specific products containing selenoneine include, but are not limited to, supplements that can substitute selenious acid and selenomethionine, prophylactic or therapeutic agents for cancers and lifestyle-related diseases such as ischemic heart diseases, and antidotes for methyl mercury.

(122) The present invention will now be described in further detail with reference to the following Examples, which are not intended to limit the present invention. The present invention may take various forms to the extent that the objectives of the present invention are achieved.

EXAMPLES

Example 1: Preparation of EgtA Expression Cassette with AsEgtA Gene Inserted

(123) (1) Extraction of Chromosomal DNA of *Aspergillus sojae* NBRC4239 Strain

(124) In a 150 ml Erlenmeyer flask, 30 mL of a polypeptone-dextrin medium (1 (w/v) % polypeptone, 2 (w/v) % dextrin, 0.5 (w/v) % KH.sub.2PO.sub.4, 0.1 (w/v) % NaNO.sub.3, 0.05 (w/v) % MgSO.sub.4.Math.7H.sub.2O, 0.1 (w/v) % casamino acid; pH 6.0) was prepared with distilled water. The medium was inoculated with the conidia of *Aspergillus sojae* NBRC4239 strain and was subjected to shake culture overnight at 30° C. The fungal cells were collected from the resulting culture medium by filtration and were placed between sheets of paper towel to remove moisture. The fungal cells were then triturated using a liquid nitrogen-chilled mortar and pestle while being chilled in liquid nitrogen. Using DNeasy Plant Mini Kit (Qiagen), the chromosomal DNA was extracted from the resulting triturated cells.

(125) (2) Preparation of Construct Plasmid

(126) The following elements were integrated into plasmid pUC19 to make a plasmid for making a construct (construct plasmid): Ptef, a promoter sequence of translation elongation factor gene tef1 (a 748 bp upstream region of tef1 gene; SEQ ID NO: 9); Talp, a terminator sequence of alkaline protease gene alp (a 800 bp downstream region of alp gene; SEQ ID NO: 10); and pyrG, a transformation marker gene that compensates for the requirement for uridine (1838 bp including a 407 bp upstream region of pyrG gene, a 896 bp coding region and a 535 bp downstream region; SEQ ID NO: 11). Specifically, the construct plasmid was prepared in the following manner.

(127) Ptef, Talp and pyrG were amplified by PCR. The PCR was performed according to the protocol provided with the enzyme using chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained above to serve as a template DNA, KOD-Plus-DNA Polymerase (Toyobo) to serve as PCR enzyme, the reagents provided with the enzyme to serve as reaction reagents, and Mastercycler gradient (Eppendorf) to serve as a PCR device. Primers used to amplify Ptef, Talp and pyrG are shown in Tables 1 to 3 below. Of the sequences shown in the tables, the sequences shown in lower case are added sequences that serve to connect the amplified fragments of Ptef, Talp and pyrG in this order and further connect them to pUC19. The amplified DNA fragments were separated in 1 (w/v) % agarose gel and purified using QIAquick Gel Extraction Kit (Qiagen).

(128) TABLE-US-00001 TABLE 1 Amplified target region Ptef Forward primer
Ptef1_-748R_pUC SEQ ID NO: 12 cgggtacccggggatcTGTG GACCAGACAGGCGCCACTC
Reverse primer Ptef1_-1R_Talp SEQ ID NO: 13 atgtactcctggtacTTTG
AAGGTGGTGC GA A C T T T G T A G

(129) TABLE-US-00002 TABLE 2 Amplified target region Talp Forward primer Talp_1F
SEQ ID NO: 14 GTACCAGGAGTACATTG GAGAGTTCTAC Reverse primer Talp_800R
SEQ ID NO: 15 CCGATCCAACCACCCGG CTATCG

(130) TABLE-US-00003 TABLE 3 Amplified target region pyrG Forward primer
PyrG_-407_F_Talp SEQ ID NO: 16 gggtggttgatcggtTGGGCT
TATTGCTATGTCCCTGAAAGG Reverse primer PyrG_143lR_pUC SEQ ID NO: 17
cgactctagaggatcCCGCACC TCAGAAGAAAAGGATGA

(131) pUC19 used was pUC19 linearized Vector provided with In-Fusion HD Cloning Kit (Clontech). Using In-Fusion HD Cloning Kit described above, the amplified Ptef, Talp and pyrG were ligated into pUC19 at In-Fusion Cloning Site located in the multiple cloning site according to the protocols provided with the kit, to obtain a construct plasmid.

(132) The resulting construct plasmid was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(133) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect fungal cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), plasmid DNA was extracted from the collected fungal cells according to the protocols provided with the kit.

(134) (3) Preparation of Construct for Inserting Gene of Interest

(135) A DNA construct consisting of gene of interest AsEgtA connected between Ptef and Talp of a

construct plasmid was prepared as follows.

(136) An inverse PCR was performed using the construct plasmid obtained above to serve as a template DNA, KOD-Plus-DNA Polymerase (Toyobo) to serve as PCR enzyme, the reagents provided with the enzyme to serve as reaction reagents, and Mastercycler gradient (Eppendorf) to serve as a PCR device. The inverse PCR was performed according to the protocol provided with the enzyme to obtain a vector fragment of the construct plasmid. Primers used are shown in Table 4 below. The amplified vector fragments were separated in 1 (w/v) % agarose gel and purified using QIAquick Gel Extraction Kit (Qiagen).

(137) TABLE-US-00004 TABLE 4 Amplified target region Construct plasmid Reverse primer Ptef₋1R SEQ ID NO: 18 TTTGAAGGTGGTGCGAACTTTGTAG Forward primer Talp₁F (above described) SEQ ID NO: 14 GTACCAGGAGTACATTGGAGAGTTCTAC

(138) To amplify the gene AsEgtA (SEQ ID NO: 1) derived from *Aspergillus sojae*, a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained above to serve as a template DNA, KOD-Plus-DNA Polymerase (Toyobo) to serve as PCR enzyme, the reagents provided with the enzyme to serve as reaction reagents, and Mastercycler gradient (Eppendorf) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify AsEgtA are shown in Table 5 below. Of the sequences shown in the table, the sequences shown in lower case are added sequences that serve to connect to the construct plasmid (between Ptef and Talp). The amplified DNA fragments were separated in 1 (w/v) % agarose gel and purified using QIAquick Gel Extraction Kit (Qiagen).

(139) TABLE-US-00005 TABLE 5 Amplified target region AsEgtA Forward primer EgtA₁F_Ptef SEQ ID NO: 19 cgcaccaccttcaaaaATGTCA CCTTTGGCTCTCTCTCC Reverse primer EgtA₂ 925R_Talp SEQ ID NO: 20 atgtactcctggtacCTAAAG ATCCCGCACCAGGCGT

(140) The vector fragments amplified as described above and AsEgtA were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct for inserting a gene of interest in which AsEgtA has been inserted. The so-obtained DNA construct consists of a DNA fragment derived from pUC19, a DNA fragment of Ptef, a DNA fragment of AsEgtA, a DNA fragment of Talp, a DNA fragment of pyrG and a DNA fragment derived from pUC19 that are connected in series in the direction from the 5' to the 3'. In other words, the DNA construct in which the sequence Ptef-AsEgtA-Talp-pyrG was inserted into the MCS of pUC19 was obtained.

(141) The resulting DNA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(142) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect fungal cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected fungal cells according to the protocols provided with the kit.

(143) A PCR was performed using the construct plasmid obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain an amplified fragment of Ptef-AsEgtA-Talp-pyrG. Primers used are shown in Table 6 below. The amplified DNA fragments were separated in 1 (w/v) % agarose gel and purified using QIAquick Gel Extraction Kit (Qiagen).

(144) TABLE-US-00006 TABLE 6 Amplified target region Ptef-AsEgtA-Talp-pyrG Forward primer Ptef₋748F SEQ ID NO: 21 TGTGGACCAGACAGGCGCCACTC Reverse primer PyrG₁₄₃₁R SEQ ID NO: 22 CCGCACCTCAGAAGAAAAGGATGA

(145) To amplify the Region 1 (Region 1 for homologous recombination) and Region 2 (Region 2

for homologous recombination) for homologous recombination to *Aspergillus sojae*, a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify the Region 1 and Region 2 for homologous recombination are shown in Tables 7 and 8 below. Of the sequences shown in the tables, the sequences shown in lower case are added sequences that serve to connect pUC19, Ptef or pyrG. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(146) TABLE-US-00007 TABLE 7 Amplified Region 1 for homologous target region recombination Forward primer RT_{-46F}_pUC SEQ ID NO: 23 cggtagccggggatcGGTTGAAGCTGTCATTGTGTGCCGA Reverse primer RT_{1766R}_Ptef SEQ ID NO: 24 cctgtctgtgtccacaAATCAG CTCCTCTGCGTGTCTCTGC

(147) TABLE-US-00008 TABLE 8 Amplified Region 2 for homologous target region recombination Forward primer RT_{1783F}_PyrG SEQ ID NO: 25 tcttctgaggtgcggACGAACTACAATGGTGCCTGGCTC Reverse primer RT_{3667R}_pUC SEQ ID NO: 26 cgactctagaggatcGCTTG GTAGAGTTGCCGCATCTGT

(148) The Region 1 for homologous recombination, Ptef-AsEgtA-Talp-pyrG and the Region 2 for homologous recombination obtained as described above and pUC19 linearized vector were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of Region 1 for homologous recombination— Ptef-AsEgtA-Talp-pyrG-Region 2 for homologous recombination has been inserted into MCS of pUC19.

(149) The resulting DNA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(150) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect fungal cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected fungal cells according to the protocols provided with the kit.

(151) To cleavage between Talp and pyrG in the construct plasmid obtained above, an inverse PCR was performed using the construct plasmid as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The inverse PCR was performed according to the protocol provided with the enzyme to obtain a vector fragment of the construct plasmid. Primers used are shown in Table 9 below. The amplified vector fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(152) TABLE-US-00009 TABLE 9 Amplified target region Construct plasmid Reverse primer Talp_{800R}(above described) SEQ ID NO: 15 CCGATCCAACCACCCGGCTATCG Forward primer PyrG_{-407F} SEQ ID NO: 27 TTGGGCTTATTGCTATGTCCCTGAAAGG

(153) To amplify the Region for looping out, a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify the Region for looping out are shown in Table 10 below. Of the sequences shown in the table, the sequences shown in lower case are added sequences that serve to connect the construct plasmid (between Talp and pyrG). The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(154) TABLE-US-00010 TABLE 10 Amplified target region Region for looping out Forward primer RT_{3731F}_Talp SEQ ID NO: 28 gggtggttgatcggACACCTAAGGCT

CCTGAGAACGGT Reverse primer RT_4771R_PyrG_-407 SEQ ID NO: 29

tagcaataagcccaaAGATAAGCTCGG TTGCGAGGGAGT

(155) The vector fragment of construct plasmid obtained as described above and the Region for looping out were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of Region 1 for homologous recombination— Ptef-AsEgtA-Talp- Region for looping out— pyrG-Region 2 for homologous recombination has been inserted into MCS of pUC19.

(156) The resulting NDA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(157) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect fungal cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected fungal cells according to the protocols provided with the kit.

(158) A part of the Region 1 for homologous recombination in the resulting plasmid DNA was removed by the following procedure. Specifically, procedures from PCR to Self-ligation were performed using the construct plasmid obtained above to serve as a template DNA with KOD-Plus-Mutagenesis Kit (Toyobo) according to the protocols provided with the kit. T100 thermal cycler (BIO RAD) was used as a device. Primers used are shown in Table 11 below.

(159) TABLE-US-00011 TABLE 11 Construct plasmid (Excluding Amplified a part of Region 1 for target region homologous recombination) Forward primer Ptef_-747F SEQ ID NO: 30 GTGGACCAGACAGGCGCCACTC Reverse primer RT_1233R_TaA SEQ ID NO: 31 TTATTGGGACGACCTGGGGTTAAGGGG

(160) The self-ligation reaction solution was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(161) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected cells according to the protocols provided with the kit.

(162) The nucleotide sequence of each DNA fragment inserted in the extracted plasmid DNA was determined to confirm that a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-AsEgtA-Talp-Region for looping out— pyrG-Region 2 for homologous recombination had been inserted was obtained.

(163) As described above, a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-AsEgtA-Talp-Region for looping out— pyrG-Region 2 for homologous recombination has been inserted into MCS of pUC19 was obtained. The resulting plasmid for DNA construct was named pEgtA_sLO_Py.

(164) A PCR was performed using the construct plasmid obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain an EgtA expression cassette. Primers used are shown in Table 12 below. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(165) TABLE-US-00012 TABLE 12 Region 1 for homologous recombination Amplified (partially removed)-Ptef-AsEgtA-Talp- target Region for looping out-pyrG-Region 2 region for homologous recombination Forward RT_33F primer CCATCCCACAAACACGGAGGAAACA SEQ ID NO: 32 Reverse RT_3649R primer ATCTGTTCTGGTTCGGAGGTGTCTGAG SEQ ID NO: 33

(166) As described above, an EgtA expression cassette in which the sequence of Region 1 for

homologous recombination (partially removed)— Ptef-AsEgtA-Talp-Region for looping out—
pyrG-Region 2 for homologous recombination has been included was obtained.

Example 2: Preparation of *Aspergillus sojae* Transformant with One Copy to Four Copies of EgtA
Expression Cassette Introduced

(167) (1) Preparation of *Aspergillus sojae* KP-del Strain

(168) An *Aspergillus sojae* KP-del strain with disruption of pyrG gene and ku70 gene on the
chromosome of an *Aspergillus sojae* NBRC4239 strain was prepared according to the description
of Example 2 in the Patent No. JP6261039.

(169) (2) Transformation of *Aspergillus sojae* KP-del Strain

(170) In a 500 ml Erlenmeyer flask, conidia of the *Aspergillus sojae* KP-del strain were inoculated
into 100 ml of a polypeptone dextrin liquid medium containing 20 mM uridine and the inoculated
medium was subjected to shake culture at 30° C. for about 20 hours. Subsequently, the cells were
collected. Protoplasts were prepared from the collected cells. The resulting protoplasts were then
transformed with 20 µg of the EgtA expression cassette using the protoplast PEG technique and the
transformants were incubated at 30° C. for 5 days or more in a Czapek-Dox minimal medium
(Difco; pH 6) containing 0.5 (w/v) % agar and 1.2 M sorbitol to obtain the *Aspergillus sojae*
transformant having the ability to form colonies.

(171) Since pyrG, a gene that compensates for the requirement for uridine/uracil, had been
introduced into the *Aspergillus sojae* transformant, the transformant was able to grow in the
uridine/uracil-free medium and was selected as strains having the introduced target gene.

(172) (3) Selection of *Aspergillus sojae* Transformant with One Copy of EgtA Expression Cassette

(173) There are eight sites on the chromosome of *Aspergillus sojae* where the EgtA expression
cassette can be introduced by homologous recombination. A strain with one copy of the EgtA
expression cassette at one of the sites was selected by PCR using the chromosomal DNA of the
Aspergillus sojae transformant extracted according to Example 1 as a template. Primers used are
shown in Table 13 below.

(174) TABLE-US-00013 TABLE 13 Amplified Insertion region of target region EgtA
expression cassette Forward primer 5_261309F SEQ ID NOs: 34-

AAGACGTGGCAAACCACTCGCA 41 RT4_-399F TTGGCTTGTTGGACCGTTGGTTGC
61_812749R TACCCAAACACCACATTCCTCGCTC 36_1378060R

AGCTTCCGAAACAATGGCGACGTG v3_18_121312R

ACTGGTACGGATAACCCATGCAGCA 60_1595128R

ATCACTACTCGTCGCAGGTGCAACAC v3_16_777681F

AAGATGATCGACCTCCCAAGGTCCCA 46_5450F AGTACAGTTGGAGTCCCTCACAGG

Reverse primer Ptef_-694R SEQ ID NO: 42 CCGGTCAAACCCAAGCAGTTGTG

(175) The strain with an EgtA expression cassette introduced due to homologous recombination at
one place was obtained. The EgtA expression cassette has been made to produce a PCR product
using one combination among the eight combinations consisting of each of eight forward primers
and one reverse primer.

(176) (4) Preparation of pyrG-Removed Strain from which pyrG Marker has been Removed in
EgtA Expression Cassette

(177) A region having the same sequence as the “Region for looping out” in the EgtA expression
cassette is common to the downstream of the eight sites that can be introduced by homologous
recombination. In the region where the EgtA expression cassette was introduced by homologous
recombination, when homologous recombination occurs between the “Region for looping out” and
the same sequence region downstream, the region comprising pyrG and Region 2 for homologous
recombination is removed by looping out. Accordingly, a strain from which pyrG was removed
from the introduced EgtA expression cassette was obtained by the following procedure.

(178) A 5-FOA-resistant strain was obtained by inoculating and culturing the conidia of the strains
obtained above, in which the EgtA expression cassette with one copy was introduced by

homologous recombination, using 20 mM uracil-supplemented Czapek-Dox agar medium containing 3 mg/ml 5-fluorouracil (5-FOA). The overview of the step from the introduction of the EgtA expression cassette to the acquisition of the 5-FOA-resistant strain is shown in the following Scheme [V]. In the Scheme [V], “up” represents the Region 1 for homologous recombination, “down 1” represents the Region 2 for homologous recombination, and “down 2” represents the Region for looping out.

(179) ##STR00005##

(180) The resulting 5-FOA-resistant strain grew on 20 mM uracil-supplemented Czapek-Dox agar medium and not on uracil-free Czapek-Dox agar medium, i.e., confirmed to show the requirement for uracil.

(181) The PCR was performed on the strain with confirmed the requirement for uracil, and it was confirmed that no pyrG-derived product was produced. Primers used are shown in Table 14 below.

(182) TABLE-US-00014 TABLE 14 Amplified target region pyrG Forward primer pyrG_8F
SEQ ID NO: 43 CCAAGTCGCAATTGACCTACAGCGCA Reverse primer pyrG_869R
SEQ ID NO: 44 ATCCCATCCCTCTTTCTGGTACCGCT

(183) The PCR was performed using a pair of primers for selection of homologous recombinant strain as shown in Table 13, and it was confirmed that the combination of primers produced PCR products was the same in the strains before and after pyrG removal

(184) The resulting pyrG-removed strain was used below as a host for further introduction of the EgtA expression cassette.

(185) (5) Preparation of *Aspergillus sojae* Transformants with Two Copies to Four Copies of EgtA Expression Cassette Introduced

(186) Strains with two copies, three copies and four copies of the EgtA expression cassettes were obtained by repeating the above steps (2) to (4); Transformation of the *Aspergillus sojae* host strain, Selection of an *Aspergillus sojae* transformant with the EgtA expression cassette and preparation of a pyrG-removed strain. The strains from which pyrG was removed by looping out were used as hosts for the next round. In addition, during the PCR for selection of homologous recombination strains, a strain in which the combination of primers producing products was newly increased by one set as compared with the host was selected.

(187) Among the strain into which 4 copies of the EgtA expression cassette were introduced, the strain that retained pyrG before removing pyrG was designated as the sA4 strain.

Example 3: Preparation of *Aspergillus sojae* Transformant with EgtA, SatA and CysB Genes Introduced

(188) (1) Preparation of SatA Expression Cassette

(189) To amplify the gene SatA (SEQ ID NO: 2) derived from *Aspergillus sojae*, a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained in Example 1 to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify SatA are shown in Table 15 below. Of the sequences shown in the table, the sequences shown in lower case are added sequences that serve to connect a vector fragment. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(190) TABLE-US-00015 TABLE 15 Amplified target region SatA Forward SatA_1F_Ptef
primer cgcaccaccttcaaaATGGGATCTATAT SEQ ID NO: 45 ACGGCGTGGC Reverse
SatA_1572R_Talp primer atgtactcctgtacTCAGCGCAAGAAT SEQ ID NO: 46
TTCTCTACTGCACC

(191) An inverse PCR was performed using the pEgtA_sLO_Py obtained in Example 1 to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The inverse PCR was performed according to the protocol provided with the enzyme to obtain a vector fragment of the

construct plasmid. Primers used are shown in Table 16 below. The amplified vector fragment was purified using QIAquick PCR Purification Kit (Qiagen).

(192) TABLE-US-00016 TABLE 16 Amplified target region pEgtA _ sLO _ Py

(Excluding AsEgtA) Forward Talp_1F(above described) primer

GTACCAGGAGTACATTGGAGAGTTCTAC SEQ ID NO: 14 Reverse Ptef_-1R(above described) primer TTTGAAGGTGGTGCGAACCTTTGTAG SEQ ID NO: 18

(193) The vector fragment amplified as described above and SatA were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-SatA-Talp-Region for looping out— pyrG-Region 2 for homologous recombination has been inserted into MCS of pUC19.

(194) The resulting DNA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(195) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected cells according to the protocols provided with the kit.

(196) The nucleotide sequence of each DNA fragment inserted in the extracted plasmid DNA was determined to confirm that a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-SatA-Talp-Region for looping out— pyrG-Region 2 for homologous recombination had been inserted was obtained.

(197) The resulting DNA construct was named pSatA_sLO_Py.

(198) (2) Bisection of SatA Expression Cassette

(199) A PCR was performed using the pSatA_sLO_Py obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain two different vector fragments of the construct plasmid. Primers used are shown in Tables 17 and 18 below. The amplified vector fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(200) TABLE-US-00017 TABLE 17 pUC19(partial)-Region 1 for Amplified homologous recombination (partially target region removed)-Ptef-SatA-Talp Forward pUC19_2238F primer GTCAATACGGGATAATACCGCGCCAC SEQ ID NO: 47 Reverse Talp_800R(above described) primer CCGATCCAACCCGGCTATCG SEQ ID NO: 15

(201) TABLE-US-00018 TABLE 18 Region for looping out-pyrG-Region 2 Amplified for homologous recombination- target region pUC19(partial) Forward RT_3731F primer ACACCTAAGGCTCCTGAGAACGGT SEQ ID NO: 48 Reverse pUC19_2252R primer TTATCCCGTATTGACGCCGGGCAAGA SEQ ID NO: 49

(202) As described above, a DNA construct (SatA expression cassette 1) in which the sequence of pUC19 (partial)-Region 1 for homologous recombination (partially removed)— Ptef-SatA-Talp was inserted and a DNA construct (SatA expression cassette 2) in which the sequence of Region for looping out— pyrG-Region 2 for homologous recombination— pUC19 (partial) was inserted were obtained.

(203) (3) Preparation of CysB Expression Cassette

(204) To amplify the gene Pgpd (promoter) derived from *Aspergillus sojae*, the gene Tamy (terminator) and the gene CysB (SEQ ID NO: 3), a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained in Example 1 to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify each gene are shown in Tables 19-

21 below. Of the sequences shown in the tables, the sequences shown in lower case are added sequences that serve to connect to pUC19, PgpD or Tamy. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(205) TABLE-US-00019 TABLE 19 Amplified target region PgpD Forward PgpD_-2000F_pUC primer cggtagcccggggacTTCACCGTATCTTAATAGAG SEQ ID NO: AACGATCCGCAAC 50 Reverse PgpD_-1R primer TGTTTGGATGTGTCTGCTGGTGTGG SEQ ID NO: 51

(206) TABLE-US-00020 TABLE 20 Amplified target region Tamy Forward Tamy_1F primer AGGGTGGAGAGTATATGATGGTACTGGT SEQ ID NO: 52 Reverse Tamy_803R_pUC primer cgactctagaggatCCATGAGATACTATGA SEQ ID NO: TATACTAAG 53

(207) TABLE-US-00021 TABLE 21 Amplified target region CysB Forward CysB_1F_PgpD primer agacacatccaaacaATGTTCCGACA SEQ ID NO: 54 AAGTATTTCGGCGCTTC Reverse CysB_1183R_Tamy primer tatactctccaccctTCAGGGGAGCA SEQ ID NO: 55 CGATGTACTTCTCCAAGT

(208) The PgpD, Tamy and CysB obtained as described above and the pUC19 linearized Vector were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of PgpD-CysB-Tamy has been inserted into MCS of pUC19.

(209) The resulting DNA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(210) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected cells according to the protocols provided with the kit.

(211) The nucleotide sequence of each DNA fragment inserted in the extracted plasmid DNA was determined to confirm that a DNA construct in which the sequence of PgpD-CysB-Tamy had been inserted was obtained.

(212) A PCR was performed using the construct plasmid obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain the amplified fragments of PgpD-CysB-Tamy. Primers used are shown in Table 22 below. Of the sequences shown in the table, the sequences shown in lower case are added sequences that serve to connect to the SatA expression cassette 1 and SatA expression cassette 2. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(213) TABLE-US-00022 TABLE 22 Amplified target region PgpD-CysB-Tamy Forward primer PgpD_-2000F_Talp SEQ ID NO: 56 gggtggttgatcggTTCACCGTATCTT AATAGAGAACGATCCGCA Reverse primer Tamy_803R_RT3745 SEQ ID NO: 57 agtagccttaggtgtCCATGAGATACTA TGATATACTAAGAT

(214) As described above, a CysB expression cassette in which the sequence of PgpD-CysB-Tamy been inserted into was obtained.

(215) (4) Preparation of SaCb Expression Cassette

(216) The SatA expression cassette 1, the SatA expression cassette 2 and the CysB expression cassette obtained as described above were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)—P_{tef}-SatA-Talp-PgpD-CysB-Tamy-Region for looping out—pyrG-Region 2 for homologous recombination has been inserted into MCS of pUC19.

(217) The resulting DNA construct was used to transform *E. coli* HST08 Premium Competent Cells (Takara Bio) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(218) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect cells. Using QIAprep Spin Miniprep Kit (Qiagen), the plasmid DNA was extracted from the collected cells according to the protocols provided with the kit.

(219) The nucleotide sequence of each DNA fragment inserted in the extracted plasmid DNA was determined to confirm that a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-SatA-Talp-Pgpd-CysB-Tamy-Region for looping out— pyrG-Region 2 for homologous recombination had been inserted was obtained.

(220) The resulting DNA construct was named pSatA_CysB_sLO_Py.

(221) A PCR was performed using the construct plasmid obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain a SaCb expression cassette. In this case, the primers shown in Table 12 were used. The amplified DNA fragments were purified using QIAEX II Gel Extraction Kit (Qiagen).

(222) As described above, a SaCb expression cassette in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-SatA-Talp-Pgpd-CysB-Tamy-Region for looping out-pyrG-Region 2 for homologous recombination been inserted into was obtained.

(223) (5) Preparation of *Aspergillus sojae* Transformant with Four Copies of EgtA Expression Cassette and One Copy of SaCb Expression Cassette

(224) The strain obtained by removing pyrG from the sA4 strain prepared in Example 2 was transformed using the SaCb expression cassette. By selecting the transformant according to the method shown in the preparation of the EgtA expression cassette-introduced strain, a strain into which one copy of the SaCb expression cassette was further introduced by homologous recombination was obtained. The strain in which pyrG before looping out was retained was named SaCb strain.

Example 4: Preparation of *Aspergillus sojae* Transformant with EgtA and MetR Genes Introduced

(225) (1) Preparation of MetR Expression Cassette

(226) To amplify the gene MetR (SEQ ID NO: 4) derived from *Aspergillus sojae*, a PCR was performed using the chromosomal DNA of *Aspergillus sojae* NBRC4239 strain obtained in Example 1 to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme. Primers used to amplify MetR are shown in Table 23 below. Of the sequences shown in the table, the sequences shown in lower case are added sequences that serve to connect to a vector fragment. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(227) TABLE-US-00023 TABLE 23 Amplified target region MetR Forward primer
MetR_1F_Ptef SEQ ID NO: 58 cgcaccaccttcaaaATGTCAGATG AGCACATCGCTCGTCA
Reverse primer MetR_1561R_Talp SEQ ID NO: 59 atgtactcctggtacCTAGTTATCG
GTGCCCACACCCTTC

(228) An inverse PCR was performed using the pEgtA_sLO_Py obtained in Example 1 to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The inverse PCR was performed according to the protocol provided with the enzyme to obtain a vector fragment of the construct plasmid. Primers used are shown in Table 16. The amplified vector fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(229) The vector fragment amplified as described above and MetR were connected using In-Fusion HD Cloning Kit according to the protocol provided with the kit to obtain a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-MetR-Talp-Region for looping out— pyrG-Region 2 for homologous recombination has been inserted into

MCS of pUC19.

(230) The resulting DNA construct was used to transform competent cells ECOS Competent *E. coli* JM109 (Nippon Gene) in accordance with the manufacturer's instructions to obtain transformed *E. coli*.

(231) The resulting transformed *E. coli* was then subjected to shake culture overnight at 37° C. in a LB liquid medium containing 50 µg/ml ampicillin. The culture medium was centrifuged to collect cells. Using FastGene Plasmid Mini Kit (Nippon Genetics), the plasmid DNA was extracted from the collected cells according to the protocols provided with the kit.

(232) The nucleotide sequence of each DNA fragment inserted in the extracted plasmid DNA was determined to confirm that a DNA construct in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-MetR-Talp-Region for looping out— pyrG-Region 2 for homologous recombination had been inserted was obtained.

(233) The resulting DNA construct was named pMetR_sLO_Py.

(234) A PCR was performed using the DNA construct obtained above to serve as a template DNA, Q5 Hot Start High-Fidelity 2× Master Mix (New England Biolabs) to serve as PCR enzyme, and T100 thermal cycler (BIO RAD) to serve as a PCR device. The PCR was performed according to the protocol provided with the enzyme to obtain a MetR expression cassette. In this case, the primers shown in Table 12 were used. The amplified DNA fragments were purified using QIAquick PCR Purification Kit (Qiagen).

(235) As described above, a MetR expression cassette in which the sequence of Region 1 for homologous recombination (partially removed)— Ptef-MetR-Talp-Region for looping out— pyrG-Region 2 for homologous recombination been inserted into was obtained.

(236) (2) Preparation of *Aspergillus sojae* Transformant with Four Copies of EgtA Expression Cassette and One Copy of MetR Expression Cassette

(237) The strain obtained by removing pyrG from the sA4 strain prepared in Example 2 was transformed using the MetR expression cassette. By selecting the transformant according to the method shown in the preparation of the EgtA expression cassette-introduced strain, a strain into which one copy of the MetR expression cassette was further introduced by homologous recombination was obtained. The strain in which pyrG before looping out was retained was named MetR strain.

Example 5: Preparation of *Aspergillus sojae* Transformant with EgtA, SatA, CysB and MetR Genes Introduced

(238) The strain obtained by removing pyrG from the SaCb strain prepared in Example 3 was transformed using the MetR expression cassette. By selecting the transformant according to the method shown in the preparation of the EgtA expression cassette-introduced strain, a strain into which one copy of the MetR expression cassette was further introduced by homologous recombination was obtained. The strain in which pyrG before looping out was retained was named SaCbMetR strain.

Example 6: Production of Selenoneine Using *Aspergillus sojae* Transformant

(239) The *Aspergillus sojae* transformants prepared in Examples 2-5 were compared for their respective abilities to produce selenoneine in the following manner. The overview of the *Aspergillus sojae* transformants used is shown in the following table 24.

(240) TABLE-US-00024 TABLE 24 Fungal No. strains Transgene (numbers) Example sA4 EgtA (4) 2 Example SaCb EgtA (4), SatA (1), CysB (1) 3 Example MetR EgtA (4), MetR (1) 4 Example SaCbMetR EgtA (4), SatA (1), CysB (1), MetR (1) 5

(241) In a 200 mL Erlenmeyer flask, conidia of each of the fungal strains were inoculated into 40 ml of a DPY liquid medium (0.5 (w/v) % histidine, 0.5 (w/v) % serine, 1 (w/v) % casein peptone, 2 (w/v) % dextrin, 0.5 (w/v) % yeast extract, 0.5 (w/v) % KH.sub.2PO.sub.4, 0.05 (w/v) % MgCl.sub.2.Math.6H.sub.2O; prepared with tap water; pH not adjusted) and the inoculated medium was subjected to shake culture at 160 rpm at 30° C. for 2 days. After the culture period,

sodium selenite was added to the culture medium to reach a final concentration of 1 mM, and the culture was further subjected to shake culture at 160 rpm at 37° C. for 4 days.

(242) After the culture period, the cell and the supernatant were separated by filtration through a Kiriya funnel (filter No. 3). The resulting supernatant was filtered through a 0.45 µm filter to obtain a supernatant fraction.

(243) On the other hand, 8 ml water was added and agitated to suspend the fungal cells and form a cell suspension. The resulting cell suspension was subjected to a heat treatment at 100° C. for 15 min. Following the heat treatment, the suspension was centrifuged to collect the extracellular fluid as the supernatant, which in turn was filtered through a 0.45 µm filter to obtain a cell fraction extract.

(244) Regarding the resulting extracts of the supernatant fraction and the cell fraction, selenoneine was quantified by LC-ICP-MS according to the conditions described in WO2017/026173

(Application No:PCT/JP2016/068128) and Yamashita et al. (THE JOURNAL OF BIOLOGICAL CHEMISTRY Vol. 285, No. 24, pp. 18134-18138, Jun. 11, 2010, "EXPERIMENTAL PROCEDURES", "Selenium Determination"). The total amount of Selenoneine and ergothioneine was subjected to HPLC analysis according to WO2017/026173 (Application No:PCT/JP2016/068128).

(245) The results of measuring the amount of selenoneine are shown in FIGS. 1A and 1B, and the results of measuring the amount of ergothioneine are shown in FIG. 2. The results of calculating the ratio of the amount of selenoneine relative to the total amount of selenoneine and ergothioneine are shown in FIG. 3 according to the results of FIGS. 1A and 2.

(246) As shown in FIGS. 1A and 1B, most of selenoneine was detected in the supernatant fraction regardless of which cell strains were used. The SaCb, MetR and SaCbMetR strains all produced a larger amount of selenoneine than the sA4 strain. In the EgtA expression cassette-introduced strain, the production of selenoneine from sodium selenite could be enhanced by strengthening the selenocysteine synthesis system and the selenate assimilation system. Similarly, potassium selenite could also be used to produce selenoneine.

(247) As shown in FIGS. 2 and 3, the SaCbMetR strain was found to produce selenoneine at high ratio. Although selenoneine and ergothioneine are difficult to isolate and purify due to their similar structures and molecular weights, these results indicated that the SaCbMetR strain can be used to obtain a high content of selenoneine by a process that simplifies the separation and purification of selenoneine.

Example 7: Toxicity of Selenium Compounds

(248) The cytotoxicity of sodium selenite was tested as follows. To quantify cell proliferation, alamarBlue Cell Viability Reagent (Thermo Fisher Scientific) was used.

(249) The mixture composed of 2× DPY liquid medium (2 (w/v) % polypeptone, 4 (w/v) % dextrin, 1 (w/v) % yeast extract, 1 (w/v) % KH.sub.2PO.sub.4, 0.1 (w/v) % MgSO.sub.4.Math.7H.sub.2O; pH not adjusted), 2× alamarBlue and 2×10^{sup.4} cells/ml the conidia of *Aspergillus sojae* NBRC4239 strain was dispensed in the volume of 100 µl into each well of a microplate. Similarly, 100 µl of an aqueous sodium selenite solution was added to each well so that the final concentration was 0 mM, 0.125 mM, 0.25 mM, 0.5 mM, 1 mM and 2 mM. The blank without the addition of the conidia and sodium selenite solution was used as a blank.

(250) After incubating the microplate at 30° C. for 30 hours, absorbance at 570 nm was measured. A plot of the value calculated by subtracting absorbance at 600 nm of the blank from the measurement result according to the instruction manual of alamarBlue reagent is shown in FIG. 4.

(251) As shown in FIG. 4, there was no growth inhibition by sodium selenite. No difference in the degree of mycelial growth was observed visually.

INDUSTRIAL APPLICABILITY

(252) The method and the transformant in one embodiment of the present invention can be used to produce a large amount of selenoneine, which is said to have antioxidant activities and the ability

to promote cell growth. Accordingly, the present invention is useful in the industrial-scale production of raw materials used to produce cosmetics and supplements with antioxidative activity.

Claims

1. A method for producing selenoneine, comprising the step of applying histidine and a selenium compound to a transformant to obtain selenoneine, wherein the transformant has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes, and wherein the transformant is a microorganism of the genus *Aspergillus* selected from the group consisting of *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchuensis*, *Aspergillus usamii*, *Aspergillus aculeatus* and *Aspergillus saitoi*.
 2. The method according to claim 1, wherein the transformant has two copies to eight copies of the EgtA gene inserted therein and can overexpress the inserted copies of the EgtA gene.
 3. The method according to claim 1, wherein the selenium compound comprises at least one selenium compound selected from the group consisting of selenic acid, selenious acid, selenium chloride, selenium tetrachloride, selenium, selenium dioxide, selenides, selenium sulfide, dimethylselenium, selenophosphate and salts thereof.
 4. The method according to claim 1, wherein the host organism of the transformant is a microorganism expressing at least one enzyme selected from the group consisting of selenic acid reductase, selenocysteine lyase and serine dehydratase.
 5. A transformant that has at least one gene selected from the group consisting of a SatA gene, a CysB gene and a MetR gene, and an EgtA gene inserted therein and can overexpress the inserted genes, wherein the transformant is a microorganism of the genus *Aspergillus* selected from the group consisting of *Aspergillus sojae*, *Aspergillus oryzae*, *Aspergillus niger*, *Aspergillus tamarii*, *Aspergillus luchuensis*, *Aspergillus usamii*, *Aspergillus aculeatus* and *Aspergillus saitoi*.
 6. The transformant according to claim 5, wherein the transformant has two copies to eight copies of the EgtA gene inserted therein and can overexpress the inserted copies of the EgtA gene.
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